

Fig. 50.1 Glomerular filtration rate (GFR) in conscious rats before and after sham operations (SO), uninephrectomy (UNX), or 5/6 nephrectomy (5/6NX). Values are mean $\pm$ standard error of the mean. (From Chamberlain RM, Shirley DG. Time course of the renal functional response to partial nephrectomy: measurements in conscious rats. *Exp Physiol.* 2007;92:251–262.)

Fig_50_1

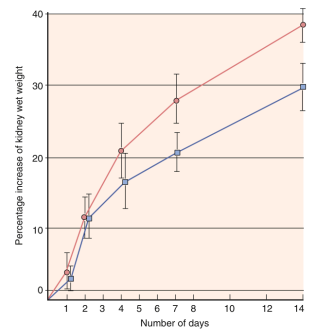


Fig. 50.2 Rate of compensatory renal growth after unilateral nephrectomy (circles) and ureter ligation (squares). (From Dicker SE, Shirley DG. Compensatory hypertrophy of the contralateral kidney after unilateral ureteral ligation. *J Physiol (Lond).* 1972;220:199–210.)

Fig_50_2

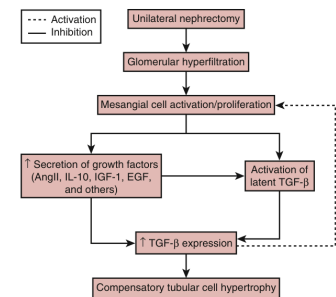


Fig. 50.3 Possible mechanisms involved in compensatory renal growth after unilateral nephrectomy. Ang II, Angiotensin II; EGF, epidermal growth factor; IGF-1, insulin-like growth factor-1; IL-1, interleukin-1; TGF-β, transforming growth factor-β. (From Sinuani I, Beberashvili I, Averbukh Z, et al. Mesangial cells initiate compensatory tubular cell hypertrophy. *Am J Nephrol.* 2010;31:326–331.)

Fig_50_3

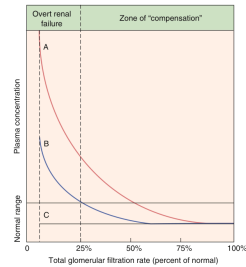


Fig. 50.4 Representative patterns of adaptation for different types of solute in body fluids with declining glomerular filtration rate (GFR). (A) For solutes cleared primarily by glomerular filtration, a rise in serum concentration occurs with each permanent reduction in GFR (e.g., creatinine). (B) For solutes that undergo substantial secretion by the tubules, a rise in serum concentration occurs only after the GFR falls below a critical value due to adaptive increases in tubular secretion (e.g., potassium). (C) For sodium, serum concentration remains normal throughout almost the entire spectrum of GFR decline due to a large capacity to reduce reabsorption by the tubule. (Modified from Bricker NJ, et al. in Brenner BM, Factor FC, eds. *The Kidney*, 2nd ed. Philadelphia: WB Saunders; 1991.)

Fig_50_4

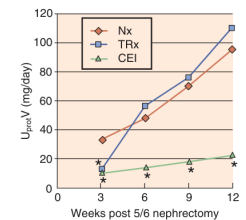


Fig. 50.7 Proteinuria levels following 5/6 nephrectomy in untreated rats (Nx) versus treatment with triple therapy (reserpine, hydralazine, and hydrochlorothiazide (TRx)). Nx + TRx, or enalapril (NX + CEI). Despite equivalent levels of blood pressure control, enalapril therapy almost completely prevented proteinuria and glomerulosclerosis, whereas triple therapy afforded no renoprotection. U_{prot}V, 24-hour urine protein excretion; *P < .05 vs. Nx at the same time point. (From Anderson S, Rennke HG, Brenner BM. Therapeutic advantage of converting enzyme inhibitors in arresting progressive renal disease associated with systemic hypertension in the rat. *J Clin Invest.* 1986;77:1993–2000.)

Fig_50_7

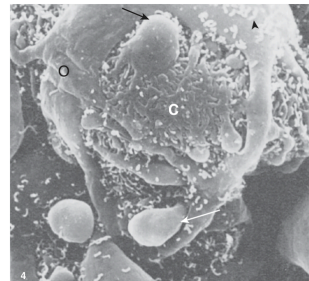


Fig. 50.5 Scanning electron micrograph of a glomerulus from a rat following 5/6 nephrectomy; view from the urinary space. Podocyte cytoplasmic blebs (arrows), numerous microvilli (arrowhead), focal obliteration (O), and coarsening (C) of foot processes are seen. ($\times 3600$) (From Hostetter TH, Olson JL, Rennke HG, et al. Hyperfiltration in remnant nephrons: a potentially adverse response to renal ablation. *Am J Physiol.* 1991;241:F85–F93.)

Fig_50_5

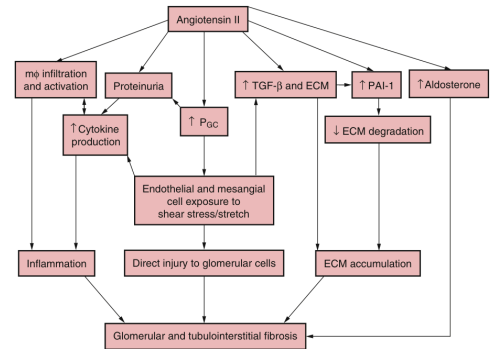


Fig. 50.8 Schematic depicting the central role of angiotensin II through hemodynamic and nonhemodynamic effects in the pathogenesis of progressive renal injury and fibrosis following nephron loss. ECM, Extracellular matrix; mφ, macrophage; PAI-1, plasminogen activator inhibitor-1; P_{GC}, glomerular capillary hydraulic pressure; TGF-β, transforming growth factor-β. (From Taal MW, Brenner BM. Renoprotective benefits of RAS inhibition: from ACE to angiotensin II antagonists. *Kidney Int.* 2000;57:1803–1817.)

Fig_50_8

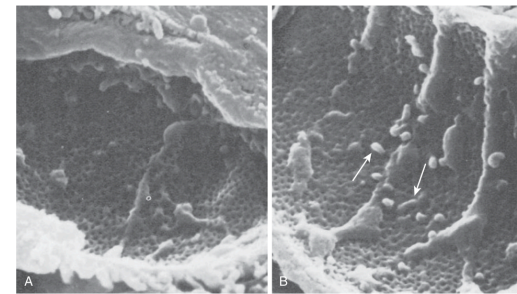


Fig. 50.6 Scanning electron micrographs of glomerular capillaries. (A) Normal endothelial appearance. (B) Rats after 5/6 nephrectomy. Scattered endothelial blebs (arrows) are often present in this group ($\times 18,000$). (From Hostetter TH, Olson JL, Rennke HG, et al. Hyperfiltration in remnant nephrons: a potentially adverse response to renal ablation. *Am J Physiol.* 1981;241:F85–F93.)

Fig_50_6

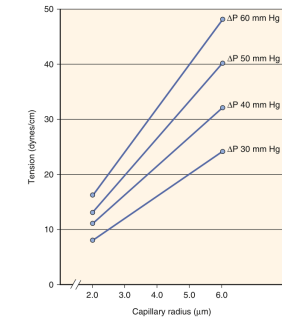


Fig. 50.9 Illustration of the synergistic effects of changes in transcapillary hydrostatic pressure difference (ΔP , mm Hg) and mean glomerular capillary radius (r) on the calculated capillary wall tension (dynes/cm). (From Bidani AK, Michael KD, Schwartz JMA, et al. Absence of progressive glomerular injury in a normotensive rat remnant kidney model. *Kidney Int.* 1990;38:28–38.)

Fig_50_9

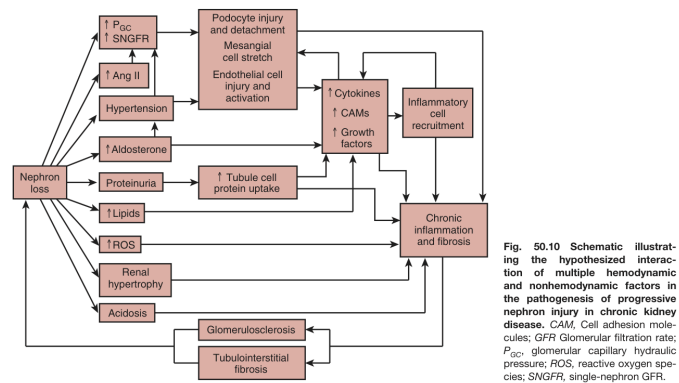


Fig. 50.10 Schematic illustrating the hypothesized interaction of multiple hemodynamic and nonhemodynamic factors in the pathogenesis of progressive nephron injury in chronic kidney disease. CAM, Cell adhesion molecules; GFR, Glomerular filtration rate; P_{GC}, glomerular capillary hydraulic pressure; ROS, reactive oxygen species; SNGFR, single-nephron GFR.

Fig_50_10

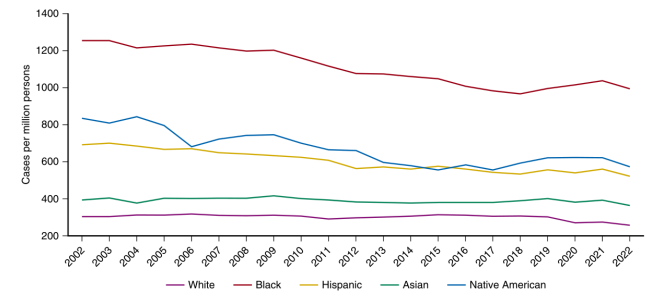


Fig. 50.11 Trends in adjusted incidence rate of end-stage kidney disease (ESKD), by race, in the U.S. population, 2002–2022. (From the U.S. Renal Data System. Annual Data Report 2024, Chapter 1: Incidence, Prevalence, Patient Characteristics, and Treatment Modalities.)

Fig_50_11

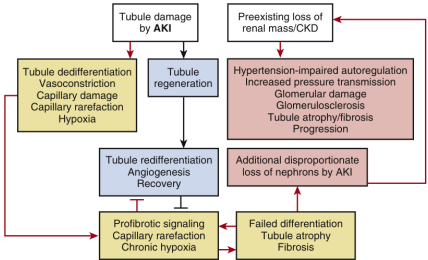


Fig. 50.12 Failed tubule differentiation and loss of renal mass after acute kidney injury (AKI) lead to hemodynamic abnormalities that cause chronic kidney disease (CKD) progression. This schematic diagram illustrates the effects of AKI that lead to tubulointerstitial fibrosis, the renal mass reduction that retards recovery of tubules regenerating after AKI, and the resulting disproportionate further reduction of renal mass that triggers hemodynamic mechanisms of renal disease progression. (From Venkatesharam MA, Weinberg JM, Kriz W, Ekdary AK. Failed tubule recovery, AKI-CKD transition, and kidney disease progression. *J Am Soc Nephrol*. 2015;26(8):1765–1776.)

Fig_50_12

Table 50.1 Hemodynamic Effects of Vasoactive Molecules Mediating Glomerular Hemodynamic Adaptations After Partial Renal Mass Ablation

Parameter	R _A	R _E	P _{GC}	Q _A	K _f	SNGFR	RPF	GFR
Angiotensin II	↑	↑↑	↑	↓	↓↔	↓↔	↓	↔
Aldosterone	↑	↑↑	↑	?	?	?	?	?
Endothelins	↑↔	↑	↑↔	↓	↓↔	↓↔	↓	↓↔
Natriuretic peptides	↓	↑(?)	↑	↔	↔	↑	↑↔	↑
Prostaglandins	↓	↓	↔	↑	↑	↑	↑	↑
Bradykinin	↓↓	↓	?	?	?	?	↑	↔
Observed changes after partial renal ablation	↓↓	↓	↑	↑	↑↓	↑	–	↓

GFR, Glomerular filtration rate; K_f, glomerular ultrafiltration coefficient; P_{GC}, glomerular capillary hydraulic pressure; Q_A, glomerular plasma flow rate; R_A, afferent arteriolar resistance; R_E, efferent arteriolar resistance; RPF, renal plasma flow; SNGFR, single-nephron GFR.

Table_50_1